

[DC] Using VR Embodiment and Co-embodiment to Improve Our Lives

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ABSTRACT

Owning another body in Virtual Reality (VR) has been proven to trigger changes in participants' social cognition, attitude, and behaviour. This PhD thesis aims to explore the capacity of VR embodiment and ways to use it to improve our lives. In study 1, we investigated embodied time travel in behaviour change to address the climate crisis and found that embodied VR experience triggered a more pronounced response in participants' perceived influence on climate action and engagement in pro-environment behaviours. Study 2 plans to explore the effect of co-embodiment (sharing an avatar with another entity) on motor skills learning. We built an initial prototype where participants could learn Tai Chi moves with an AI teacher, and plan to investigate whether co-embodiment could enhance learning.

Index Terms: Embodiment, Co-embodiment, Shared agency, Behavioural change, Motor skill learning, Taichi, Sustainability

1 INTRODUCTION

Recent advancement in technology, such as 5G and AI, has significantly enhanced various aspects of life, but not without adverse effects. The surge in energy consumption has exacerbated our climate crisis, while the increasing prevalence of sedentary, knowledge-intensive tasks is generating new challenges for our physical and mental health. In this PhD thesis, we try to address the overall research question: **how can technology be utilized as a tool to promote sustainable behaviours and foster learning, ultimately enhancing individuals' cognitive and physical capabilities?** This thesis explores various way in which we can ensure that technological advancements do not come at the expense of the environment and human well-being, through Virtual reality (VR) technology. In particular, we explore the phenomenon where participants take 1st person perspective of the experience, which has been proven to influence their attitudes [1, 3] and improve motor skill acquisition [4]. By linking cognitive and emotional engagement with procedural learning, this research aims to demonstrate the broader impact of embodied experiences on human behavior.

2 STUDY 1: EMBODIMENT AND CLIMATE CHANGE

Our first research question (**RQ1**) was: can embodied experiences in VR influence individuals' attitudes and behaviours toward climate change, as tools to promote sustainable behaviours, ultimately contributing to a better future? A total of 41 participants were divided into two groups: an embodied group that interacted with a virtual environment through full-body avatars, and a non-embodied group that observed the scenarios from an invisible observer's point of view [5] (Figure 1). The VR simulation showed how climate change affects three generations of one family progressively, making abstract and distant concept of climate change a tangible and personal issue. The final scene presented an optimistic scenario of a future where humans had successfully combated climate change through collective action. The evidence suggests that there is an effect of the scenario on the carbon footprint response, even 6 weeks after

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Figure 1: The effect of embodied experience on behavioural change towards climate change in the previous study.

the VR exposure, irrespective of condition. Additionally, increases were found in participants' perceived influence on climate action and engagement in pro-environmental behaviors, with the embodied group showing a more pronounced response in the short term. These findings suggested that immersive VR experiences that incorporate virtual embodiment can be an effective tool in enhancing awareness and motivating pro-environmental behavior by providing a powerful and personal perspective on the impacts of climate change.

The results suggested that embodiment strengthens the sense of personal responsibility and engagement in climate action. These findings inform the design of Study 2, where we explore whether mechanisms such as perspective-taking and agency could enhance motor skill learning in a different context. Given that embodiment can shape our behaviours and cognition, it is worth to investigate how to use it to optimize physical learning outcomes.

3 STUDY 2: CO-EMBODIMENT FOR MOTOR SKILL LEARNING

Unlike the broad and sometimes intangible goals of behavioral change, motor skill learning focuses on tangible, measurable outcomes, such as mastering Tai Chi forms, illustrating a natural progression from cognitive understanding to physical skill acquisition. Many researchers explored learning Tai Chi in VR: Tian et al. found VR to be the most effective for training accuracy, while Mixed Reality offered better social experiences [6]; Yan et al. introduced TaoRelax, a VR relaxation system using electromyography (EMG) for feedback, showing reduced psychological stress and improved meditation abilities [8]. However, to date, no study has investigated Tai Chi learning from a first-person perspective (1PP)—an approach feasible only in VR. Recent studies have highlighted the impact of shared agency and co-embodiment on motor skill learning [4]. For example, Kodama et al. used a 1PP approach where learners shared control of a virtual body with a teacher during dual tasks. Their results showed that controlling one avatar with a teacher based on co-embodiment improved motor skill learning efficiency compared to merely seeing the teacher's perspective sharing or learning independently. However, their study only examined shared agency with a real teacher and simpler tasks, leaving a gap in exploring AI teachers and more complex motor activities.

We propose using Tai Chi, a health-promoting and motor task enabled by motion capture (mocap) and machine learning (ML) [2, 7], to study shared agency and co-embodiment in VR for motor skill learning. We will focus on the development and evaluation of an AI-driven Tai Chi teacher (key innovations see Figure 2).

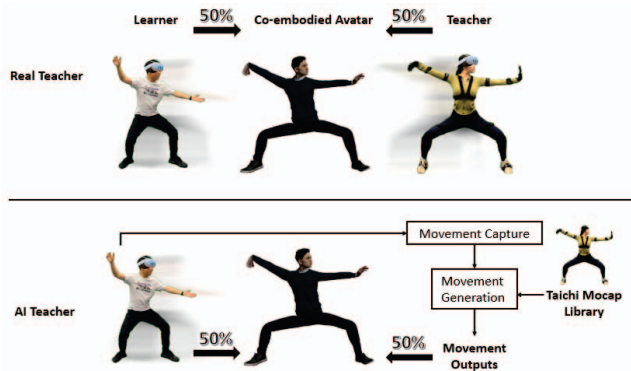


Figure 2: **Study 2 - Upper: Co-Embodiment with a Real Teacher:** learner's avatar driven by weighted average of their own movements and the Teacher's. **Lower: Co-Embodiment with an AI Teacher:** learner's avatar drive by weighted average of their movements and AI (the AI Teacher's movement generated in real-time by an Interaction ML Model responding to the learner's movements).

Shared Agency System: A networked system will be developed to enable participants to train collaboratively with either real or AI teachers. The system will capture the real-time movements of both the participant and the teacher as input. By applying a shared control equation, inspired by Kodama et al., the system will assign equal weighting (50% each) to the inputs from the participant and the teacher [4]. These combined inputs will control the movement of a virtual avatar through an inverse kinematics system, such as Final IK¹, allowing participants to view their optimized Tai Chi movements in real-time from a 1PP as their own body.

AI Tai Chi Teacher: Compared with real-time network control of the body by real teachers, an AI-driven teacher will be developed Using real-time motion input from participants, the system will employ gesture recognition models, such as Gesture Follower [2], within Unity to compare participant movements against a pre-recorded database of expert Tai Chi routines captured with the Perception Neuron system. The AI will identify the most optimal movements corresponding to the participant's position and seamlessly integrate these with the participant's input within the shared agency system. This will provide precise and adaptive guidance, enabling high-quality instruction without the need for human resources.

Comparative Analysis: The effectiveness of Tai Chi training will be evaluated by comparing outcomes across real and AI teachers. This comparative analysis addresses our (RQ2): Can co-embodiment with AI-generated movement achieve Tai Chi training outcomes comparable to those of co-embodiment with a real teacher, thereby supporting sustainable physical activities?

We developed a VR prototype using MoCap to record Tai Chi movements from professional dancers, which we presented to learners in 3PP. Using Mivry and InteractML², we trained ML models in Unity to provide real-time feedback. A pilot study showed participants favored VR over video-based learning for its convenience, 3D visualization, and real-time error correction. However, current models lack precision in motion recognition. To improve, we plan to integrate efficient feedback systems like Gesture Follower and develop faster, smaller models.

4 CONCLUSIONS

This doctoral research explores the application of embodied VR experiences for sustainable behavior change and motor skill learning, focusing on Tai Chi as a complex motor task. By leveraging

virtual co-embodiment, it explores both cognitive and emotional engagement and procedural and physical skill acquisition. The study integrates MoCap, ML, and AI-driven teachers to address gaps such as the efficacy of AI compared to human teachers and the role of shared agency in learning. Key contributions include the development of an AI Tai Chi teacher capable of providing adaptive feedback and investigating the impact of shared agency on skill acquisition from a first-person perspective. The research bridges cognitive learning and physical skill mastery by combining motion capture, machine learning, and real-time feedback, advancing VR-based education and training. It emphasizes measurable outcomes such as skill improvement and user engagement, ensuring both academic and practical relevance.

Further exploration will refine AI-driven feedback mechanisms, compare learning outcomes between real and AI teachers to inform hybrid approaches and scale the system for diverse user groups while addressing ethical considerations. Participation in the doctoral consortium aims to refine these aspects, foster interdisciplinary collaboration, and explore broader implications for VR, AI, and education. These two studies contribute to a broader understanding of how VR can facilitate cognitive and behavioral change, either through personal avatars (Study 1) or shared agency with AI (Study 2). This research aims to optimize VR for behavior change and skill acquisition, highlighting its potential for sustainable learning and personal development.

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¹<http://root-motion.com/>

²<https://www.marui-plugin.com/mivry/>, <https://interactml.com/>